



In vivo effects of two shoulder girdle motor control exercises on acromiohumeral and coracohumeral distances in healthy men

Fernanda B. Charry^a, María Jesús L. Martínez^a, Liliana Rozo^b, Fernando Jurgensen^c and Juan Guerrero-Henriquez^d 

^aManual Therapy Master's Degree, Faculty of Rehabilitation Sciences, Andrés Bello University, Santiago, Chile; ^bPostgraduate Department of Manual Therapy, Faculty of Rehabilitation Sciences, Andrés Bello University, Santiago, Chile; ^cClinical Applicationist of Ultrasound for General Electric Company, Hoser Ingeniería, Santiago, Chile; ^dRehabilitation and Human Movement Sciences Department, Faculty of Health Sciences, University of Antofagasta, Antofagasta, Chile

ABSTRACT

Introduction: Altered motor control and proprioceptive deficits are associated with kinematics dysfunctions and may cause alterations in subacromial space (SAS) that could lead to shoulder pathologies. Dimensions of the subacromial space, as well as interventions aimed at its normal restitution, can be explored by ultrasound (US).

Objective: To describe the effect of two shoulder girdle motor control exercises with cognitive training strategies on SAS dimensions, measured with US.

Methods: Cognitive movement control strategies, with visual and haptic feedback were applied on 21 healthy participants. SAS dimensions were measured through *in vivo* variations of acromiohumeral (AHD) and coracohumeral distances (CHD) using US.

Results: Our results show that as exercise repetitions are performed, an increasing trend in both measures can be observed, being wider for AHD (i.e. humeral head descent exercise) than CHD (i.e. scapular retraction exercise).

Conclusion: Specific cognitive and motor control exercises improve congruence joint and centering of the humeral head.

KEYWORDS

Motor control;
acromiohumeral distance;
coracohumeral distance;
ultrasonography and
shoulder

Introduction

Motor control can be defined as the ability of the central nervous system to execute a concrete and specific movement in a more precise, functional, and coordinated way by means of feedforward and/or feedback cues, to allow a more leisurely and conscious movement using sensory and proprioceptive information in forms of verbal, visual or haptic inputs [1–4]. In the last 25 years, the relationship between sensorimotor system and musculoskeletal disorders has been established [5,6] and the importance of considering this relation in shoulder girdle sensorimotor retraining within the management of patients with shoulder conditions [7].

Altered motor control and proprioceptive deficits are associated with kinematic dysfunctions in the shoulder complex [8–10], which can lead to changes in rotation axis and abnormal humeral head translation into the glenoid cavity during shoulder elevation [11,12], causing alterations in dimensions of the subacromial space [13,14], leading to rotator cuff pathologies (RCP) [15,16]. These shoulder pathologies are related to deficits in muscle activation, alteration in movement patterns, and proprioceptive mechanisms

of the glenohumeral joint [17] and scapula [18,19] and also are one of the most common causes of referrals to physical therapy [20].

Subacromial space measures can be estimated using ultrasound (US) devices. US is a noninvasive imaging method that have demonstrated their usefulness in dynamic exams of the musculoskeletal system for their accuracy, reliability and low cost in the diagnosis of the shoulder complex [21,22]. Commonly used variables for description of subacromial space are acromiohumeral distance (AHD) [43,24] and coracohumeral distance (CHD) [23,25]. AHD and CHD have been used for assessing changes in subacromial space after glenohumeral and scapular exercises [26,27] and have been associated to glenohumeral pathologies [28,29].

Motor control exercises and neuromuscular (re) learning, proprioceptive and functional training are recommended for rehabilitation in the context of symptoms related to shoulder impingement syndrome (SIS) [6,30–32]. Previous studies have shown improvement in disability, functionality and pain in patients with RCP and SIS [20,24,30] using this type of specific exercises. Furthermore, in the medium and long term, it has been reported that this exercise strategy can reduce the need for arthroscopic subacromial

decompression [33], as well as motor control exercises used in the clinical setting could be performed to improve arthrokinematic dysfunctions such as excessive humeral head ascension (with humeral head descent exercises) and humeral head anteriorization (with scapular retraction exercises) [18,34].

Moreover, different cognitive retraining strategies that influence motor learning in patients while executing a specific movement pattern can lead to better results during the treatment [31]. Consequently, focusing the rehabilitation on restoring the functionality of the patients from a cognitive and perceptual approach seems to be important to restore the neuromuscular disorders of the shoulder complex. To the best of our knowledge, variations in subacromial space have not been evidenced in real time when applying exercises involving both motor control and cognitive training strategies.

For this reason, the aim of this study was to describe the effect of two shoulder girdle motor control exercises with cognitive training strategies on AHD and CHD, measured with US. It was hypothesized that the humeral head descent exercise increases the AHD, and the scapular retraction exercise increases the CHD.

Materials and methods

A quasi-experimental with one group pretest-during test design was used in this study. All participants read and signed an informed consent for this study. The protocol for this investigation was approved by the ethics committee of the Faculty of Rehabilitation Sciences, Andrés Bello University, Chile.

Participants

A total of 21 healthy pain-free male participants were enrolled using sample size calculation (further details in statistical analysis subsection). Inclusion criteria consisted in no history of trauma or condition in the spine or dominant upper limb. All participants were evaluated in a single session. Prior to participants enrollment, an initial evaluation consisting in seven orthopedic tests was conducted to rule out RCP conditions: painful arc test for screening for glenohumeral joint; Neer Test, Hawkin–Kennedy Test, Yocum Test, Sulcus Test, Anterior Drawer test and Jobes Relocation test [11,35,36].

Protocol

Figure 1 summarizes the study protocol. Three physical therapists (PT) and one medical technologist applied the investigation protocol. PT-A and PT-B oversaw initial assessment of participants and motor control exercise education, respectively. Both had a master's degree in Orthopedic Manual Therapy, with post-

graduate education on musculoskeletal area and active participation over 8 years on clinical experience on musculoskeletal and sport rehabilitation research. After initial assessment and education, PT-C oversaw motor control exercises execution while medical technologist registered US images. PT-C has a master's degree in Orthopedic Manual Therapy and had a master's degree in Orthopedic Manual Therapy and had more than 20 years in musculoskeletal academic and clinical experience. The medical technologist, who oversaw US imaging, has more than 20 years of experience with a major in radiology and medical physics and a bachelor in Radiological Sciences and certification in upper and lower limb musculoskeletal US.

Motor control exercises

Cognitive movement control strategies (verbal, visual, and haptic feedback) were chosen for this study and two motor control exercises: (1) humeral head descent exercise and (2) scapular retraction exercise. Table 1 gives detailed information regarding exercises.

Ultrasound measurements

A US device (General Electric, Logiq e pro model) was used with a linear transducer model 12 L-RS of 192 elements which have a footprint of 12.7×47.1 mm. A 10 MHz average frequency was used. The imaging protocol used was the general musculoskeletal protocol proposed by US manufacturer. Participants were instructed to remain seated with their feet flat on the ground while their upper extremities were positioned in the scapular plane with 70° of abduction, since it allows a better US view of AHD and CHD. A manual goniometer was used to control testing position.

AHD distance was defined as the tangential distance between the humeral head and the edge of the acromion [37]. CHD was defined as the tangential distance between the coracoid and the humeral head in a longitudinal sonogram [38]. AHD was used for the exercise of descent of the humeral head while CHD was used for the scapular retraction exercise. Both AHD and CHD were measured in millimeters (mm).

An initial measurement was taken in the initial position of each exercise (i.e. abduction in scapular plane at 70°). During exercise execution, AHD and CHD distance were measured 5 times (on each repetition) with 10 seconds rest intervals between repetitions (Figure 2).

Statistical analysis

The sample size for this study was calculated on G*Power version 3.1.9.4 [39] based on the formula for comparing two means and a priori distances in mm reported in previous studies [11,40]. A 95% confidence

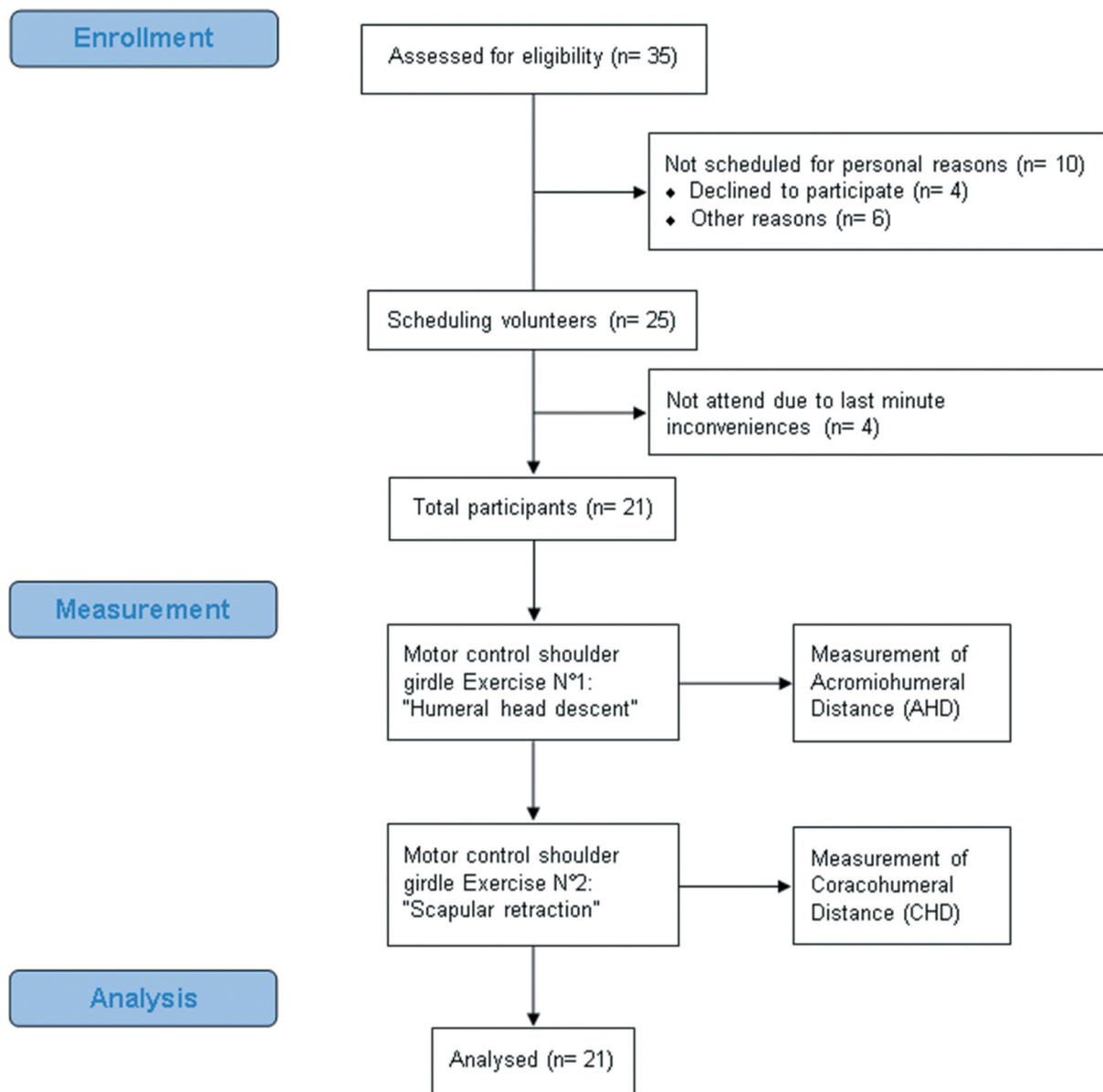


Figure 1. Protocol flow diagram.

level, 90% statistical power, a precision of 0.94 mm, a variance of 0.90 were selected for calculation, resulting in a sample size of 12 participants to detect significant differences. Notwithstanding the foregoing, we included additional participants in case of erroneous readings by the US. Further statistical analysis was carried out in the SPSS v25.0 program (IBM Corp., Armonk, NY). Demographics are presented as mean and standard deviation. The Shapiro Wilk statistical test was applied to assess data distribution. For normal distributions, data were expressed as mean standard deviation and for data with non-normal distribution medians and 25% and 75% percentiles were selected. For categorical variables, absolute frequency and percentage were used for data presentation.

A reliability analysis was conducted using the intra-class correlation coefficient (ICC) and its respective confidence interval [lower bound – upper bound] on the AHD and CHD measurements obtained, where a value

lower than 0.5 was indicative of low reliability, values between 0.5 and 0.75 were indicative of moderate reliability, values between 0.75 and 0.9 as good reliability and values higher than 0.9 as excellent reliability [41].

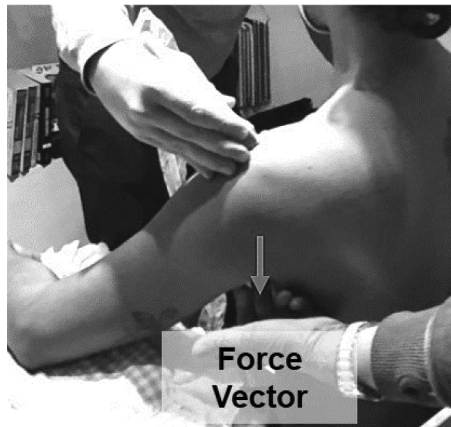
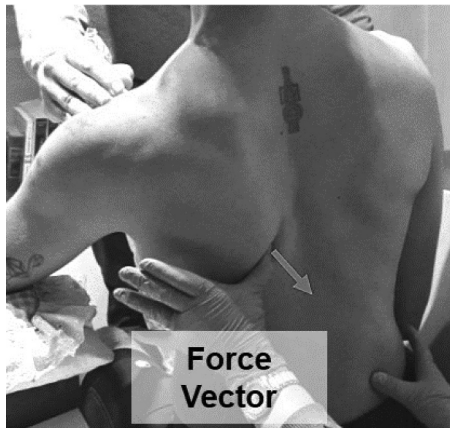
Normalization of AHD and CHD was carried out to quantify the rate of change (Δ AHD and Δ CHD, respectively) in each repetition, using as base parameter the initial positioning of each exercise (i.e. scapular plane at 70° of abduction).

Results

Demographic data is summarized in Table 2.

Data expressed as mean $\pm$ standard deviation and absolute frequency (absolute frequency percentage, %).

Table 1. Cognitive movement control strategies and motor control exercises.

Cognitive Movement Control Procedure			
Strategies Used	• Visualization: The anatomy of the shoulder girdle (bones-muscles) was visually displayed using a 3D anatomy application. • Haptic feedback: The physical therapist palpated the scapula and shoulder of each participant. Then, each participant felt with his own hands. • Visual feedback: Two videos were recorded with each of the exercises to show and visualize how the movements looked like. • Verbal command: A clear, short, and precise verbal instruction was given of the movement to be executed in each of the exercises.		
Motor and Cognitive Learning	• Slow, conscious, and low effort repetitions of each of the exercises were requested. • Each exercise was practiced with verbal, haptic, and visual feedback facilitation. Five repetitions were performed pre-measurement.		
Measurement	• Each exercise was performed with verbal and haptic facilitation. No visual feedback. • The exercise was repeated 5 times, with a 10-second rest between each repetition. Each repetition was hold by 5 seconds. • The participants were in a seating position without backrest, with their feet resting on the floor. Upper limb measured in scapular plane with 70 ° shoulder abduction.		
Motor Control Exercises			
(1) Humeral Head Descent Exercise		(1) Scapular Retraction Exercise	
Objective	Recruitment of depressor muscles of the humeral head (supraspinatus, infraspinatus, teres minor, subscapularis, and long head biceps).	Objective	Recruitment stabilizing muscles of the scapula (serratus anterior, middle and lower trapezius portions, rhomboids, and pectoralis minor).
Outcome	Inferior gliding of the humeral head through the acromiohumeral distance on ultrasound.	Outcome	Posterior gliding of the humeral head through the coracohumeral distance on ultrasound.
Implementation	- The physical therapist palpated the humeral head with her index finger, middle finger, and ring finger below the shoulder (at the axillary level) of the participant. - The participant, without moving the arm, had to make a downward pressure with the shoulder on the physical therapist fingers for 5 seconds.	Implementation	- The physical therapist palpated the inferior angle of the scapula and simultaneously palpated the participant's contralateral hip. - The participant, without moving the arm, had to bring the lower angle of his scapula toward the contralateral hip for 5 seconds. To do this, the physical therapist palpated these two points.
Verbal Command	'Without moving your arm, imagine that with your humeral head you want to push down against the fingers of the physical therapist with your maximum force'.	Verbal Command	'Without moving the arm, push the physical therapist's fingers with your scapula with your maximum force, bringing the shoulder to the opposite hip'.
			

Reliability analysis

ICC values for AHD were 0.94 [0.92–0.95] and for CHD were 0.96 [0.95–0.97].

Acromiohumeral and coracohumeral differences

Table 3 summarizes mean AHD and CHD across repetitions. Figure 3 illustrates the variation across repetitions of normalized AHD and CHD. An increasing trend in both measures can be observed, being wider for AHD (i.e. humeral head descent exercise) than CHD (i.e. scapular retraction exercise).

Data expressed as mean $\pm$ standard deviation.

Discussion

The aim of this study was to describe the effect of two shoulder girdle motor control exercises with cognitive training strategies on AHD and CHD, measured with US. Our results suggest that these exercises, using the proposed cognitive training strategies, may increase subacromial space dimensions.

Our findings regarding humeral head descent exercise and its relationship with the AHD agreed with the findings of 32, who suggests that isometric exercises are associated with an increase in the AHD, indicating that this exercise favors the inferior translation of the humeral head by consciously activating the rotator cuff muscles [32]. Furthermore, in a quasi-experimental study conducted by 11, reported the use of two different dynamic neuromuscular control exercises (descent and posterior exercises of the humeral head) for the

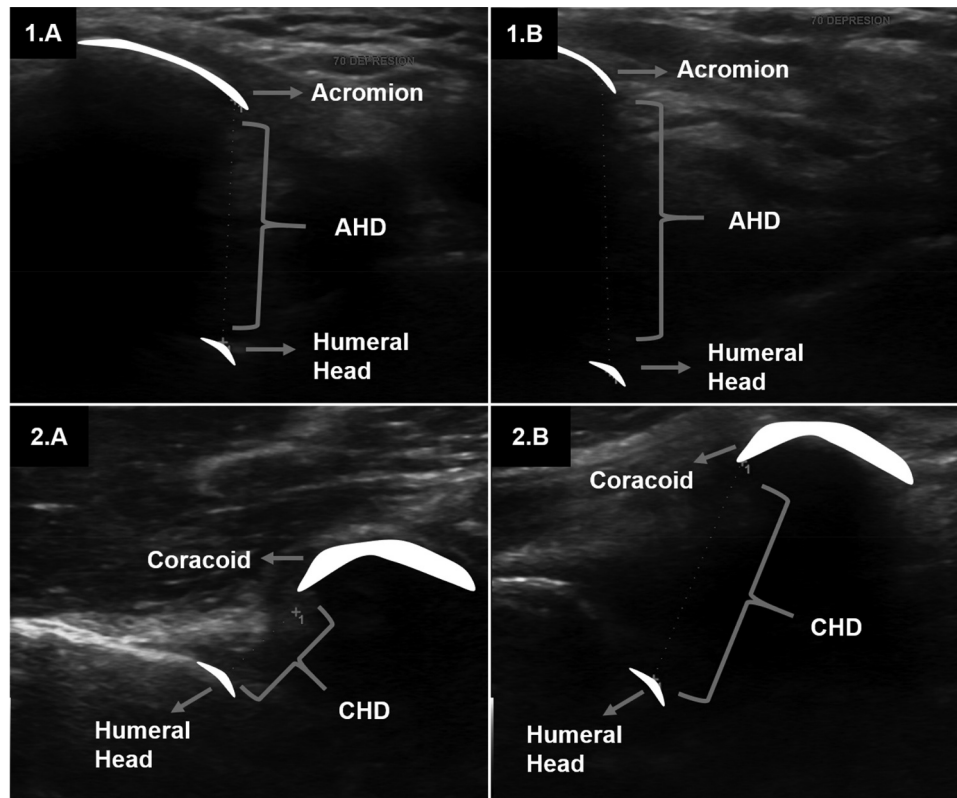


Figure 2. Ultrasound references are used to mark and calculate the acromiohumeral distance (AHD) (1.A and 1.B) and the coracohumeral distance (CHD) (2.A and 2.B). 1.A is on the participant's starting position and 1.B during the maximal effort to press the humeral head descent exercise. 2.A and 2.B are the same measurement steps, but for the coracohumeral distance (CHD), performing scapular retraction exercise.

AHD and for the anterior-posterior distance, demonstrating the increase shown on AHD with those exer-

be reliable, since a measurement lower than this could indicate association with RCP [28]. Moreover, our ICC values indicated excellent reliability for both measurements, therefore, results obtained in this study are reliable for comparative purposes.

Additionally, a better visibility of the CHD was achieved when positioning participants in 70° of abduction in a scapular plane. This can be considered an option to reduce mechanical compression between the coracoid, humeral head, and adjacent soft tissues and allows working both humeral and scapular stabilizing muscles. Thus, an interesting treatment proposal arises with the concept of motor control training with different sensory inputs, specifically for alterations of the subscapularis tendon and long head of the biceps. As stated, the most affected space in patients with pathologies in these tendons is the CHD [25,28,44].

As illustrated in Figure 3, AHD and CHD measures progressively increase with the repetitions. This increase may be due in part by a correct muscle recruitment, associated with visual, cognitive, and haptic learning (cognitive and motor control) done prior and during exercise execution. This approach, as pointed out by 31, is an important advance toward recognizing individual differences of each patient and highlights the need to adapt interventions based on evaluation, reaffirming the importance of including this type of cognitive commands at the time of teaching and perform a specific exercise [31]. Furthermore,

Table 2. Demographic characteristics of participants.

Variable	Participants (n = 21)
Age (year)	22.95 ± 1.40
Height (cm)	1.78 ± 0.05
Weight (kg)	77.67 ± 9.29
Dominance	16 (76.1)
Right	5 (24.9)
Left	
Shoulder measured	5 (23.8)
Right	16 (76.2)
Left	

Table 3. Base and between exercise repetition differences in acromiohumeral (AHD) and coracohumeral distance (CHD).

Repetition	AHD (mm)	CHD (mm)
Base	11.06 ± 3.92	20.79 ± 4.48
1st Repetition	13.58 ± 3.41	25.01 ± 5.94
2nd Repetition	14.82 ± 4.48	25.65 ± 5.29
3rd Repetition	15.17 ± 4.06	27.14 ± 6.81
4th Repetition	16.26 ± 5.19	27.58 ± 6.98
5th Repetition	16.53 ± 5.99	26.46 ± 6.48

cises [11]. Regarding the scapular retraction exercise and its relationship with the CHD, our results were higher than previous studies. 42, found a CHD ranging 10.5–16.1 mm and 43, with 10.3 mm [23,42]. Considering these results and the ones obtained by 28, a CHD greater than 10 mm in healthy subjects can

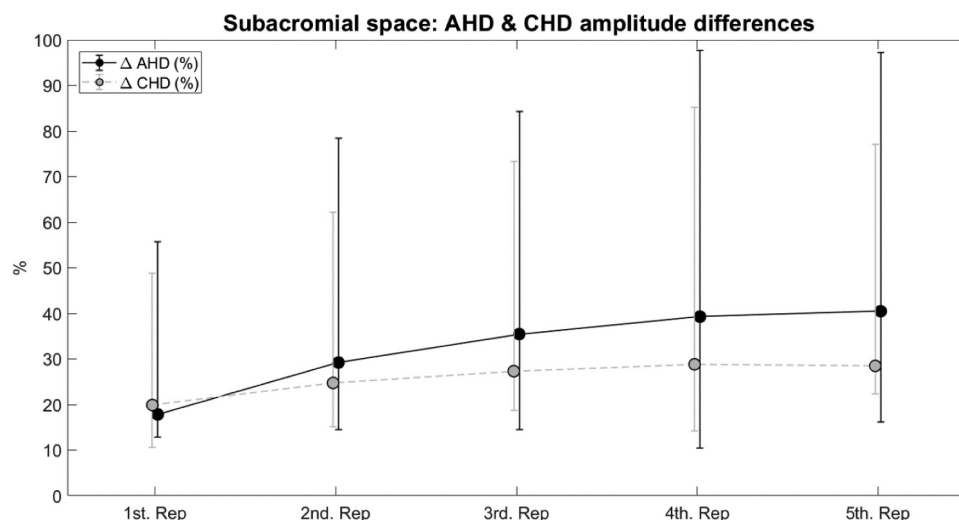


Figure 3. Normalized acromiohumeral (Δ AHD; black dots and line) and coracohumeral distances [Δ CHD; gray dots and dashed line] between humeral head descent and scapular retraction exercises repetitions. Data is expressed as median and 25th and 75th percentiles.

contemporary neuroscience and motor control theories [45] argue that pain alters motor pattern/control in different ways in response to the individual's conscious and unconscious perception of threat, and this can generate changes in motor function and movement. Moreover, characteristics of muscle synergies can change in the presence of pain [46] and they are associated with biomechanical, motor control, and performance variables [47].

Exercises analyzed in our research support the idea of including dynamic exercises of the rotator cuff muscles in humeral centering, through modification of subacromial space dimensions, and therefore improve shoulder function and decrease the subacromial pain [11,12,15,33,37]. In a randomized clinical trial, the effectiveness of dynamic humerus centering exercises in RCP was analyzed, stating that these exercises can reduce shoulder pain in subjects with SIS [20]. Furthermore, 48, conducted a systematic review and concluded that there is consistent information that supports the importance of involving the activation of the scapulohumeral musculature in patients with SIS, which is also related to the results of this study and the possibility of including cognitive and motor exercises for rehabilitation in RCP. Therefore, shoulder girdle lowering, and retraction exercises may have benefits in the prevention and treatment of RCP and other disorders related to the shoulder joint [48].

Another point to discuss is whether motor control exercises are superior to other therapeutic modalities. Dos Santos, et al. reported that shoulder neuromuscular exercise programs that incorporates strength, coordination, balance, proprioception, motor control, and functional kinetic chain work were superior to a standard care exercise program that emphasizes strength training to increase muscle mass in patients with traumatic shoulder instability [49]. On the other hand,

strength-based exercise programs can directly affect the functional capacity of dynamic shoulder stabilizers, which by improving their performance can impact on joint stability and proprioception [50,51]. 50, found an improvement in glenohumeral joint repositioning after performing a strengthening exercise program without varying their intensity, while better neuromuscular control of the shoulder was observed in the group that performed exercises with different intensities. These differences are due partly to the improvement in the sensitivity of muscle spindles, confirming that stimulating the muscle spindle, as well as the different structures active and passive shoulder, in different ways through various treatment strategies positively impact the outcomes involved in shoulder dysfunction such as pain [19], decreased function [7], motor control, stability, muscle weakness [52], proprioception and alteration in mobility [49]. Other studies based on motor control exercise programs in SIS and RCP use biofeedback and electromyography with controversial results. 53, demonstrated that using electromyographic biofeedback improved motor control in asymptomatic and symptomatic subjects, while 54, found no difference in pain and function [53,54]. As stated, motor control exercises should be included in the management of patients with RCP and SIS, considering a multidimensional perspective, rather than a single approach.

Physical therapists in clinical practice often teach several exercises for RCP, however, the proper understanding and execution of these exercises can be particularly difficult for some individuals, since they require a complex and atypical voluntary neuromuscular activation sequence [11]. According to this background, different cognitive movement control strategies are included in this study, so the participant can get optimal performance at both cognitive and motor levels during

the execution of the exercise of descent and retraction of the shoulder girdle. The use of US in real time can be an alternative to complement the verbal and palpatory instructions during the execution of the movement. Besides the education phase of the exercises, visual feedback collaborates with exercise familiarization, accompanied by a previous anatomical visualization. Additionally, these strategies can help movement specialists to get a better understanding of clinical findings of movement evaluation, treatment guidance, ease interdisciplinary communication, and promote a modern pathophysiological approach within a multidimensional perspective since the clinical reasoning skills of the manual therapist are essential [31,55].

Limitations

We must state several limitations. Our results correctly describe an *in vivo* modification of subacromial space during the execution of the exercises, but it does not allow to determine the effect of these two exercises in the long term, as in a preventive rehabilitation program or a prospective cohort on the treatment of RCP. On the other hand, considering that the collected sample satisfied sample size calculation carried out, it is important to note that this study was during the global context of COVID-19 and even though the safety of each of the participants was protected, this context limited the possibility of taking a bigger data sample or population comparisons. Moreover, healthy pain-free males were selected, so findings cannot be extrapolated to individuals with shoulder mobility restrictions or shoulder pathologies. For future studies, it is recommended to increase the number of the total sample to have a better representation and statistical power at a national level and maybe complement with the use of surface electromyography, being able to determine if there is any type of correlation between the increase of both anatomical spaces (AHD and CHD) and increased muscle activity using the proposed cognitive and motor control exercises in musculoskeletal dysfunctions.

Conclusion

We can conclude that, through an *in vivo* exploration, specific cognitive and motor control exercises improve congruence joint and centering of the humeral head, through a progressive increase in AHD and CHD which favors the displacement of the humeral head.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

Notes on contributors

Fernanda Charry, PT, MSc, has 8 years of experience in rehabilitation area, graduated from Manuela Beltrán University. She is certified in Alternative Therapies, Traditional Chinese Medicine and Clinical Acupuncture at the School of Acupuncture, Moxibustion and Tuina from Beijing University of Chinese Medicine in Beijing, China. She is also certified as Pilates instructor at Balanced Body, Colombia-USA. She has a master's degree in Orthopedic Manual Therapy from the Andrés Bello University and is an active member of the Colombian College of Physical Therapists (COLFI), Colombian Association of Physical Therapy (ASCOFI) and currently a member of RIG-TMO Colombia, Colombian Scientific Association of Orthopedic Manual/Manipulative Orthopedic Therapy.

Maria Jesus Martínez León, PT, MSc, has a master's degree in Orthopedic Manual Therapy, with post grade education in musculoskeletal area and active participation on collaborative research with clinical experience in musculoskeletal and sport rehabilitation.

Liliana Rozo, PT, MSc, currently works at the Master of Orthopedic Manual Therapy Program in Andrés Bello University in Santiago, Chile. She completed the Certification Specialty in Orthopedic Manual Therapy at St. Augustine University, FL, USA. She has a master's degree in Manual Therapy from Andrés Bello University, Chile. His research interests include limb dysfunctions and motor control with the integrative approach to manual therapy and clinical reasoning. She is active in COLKINE Chile, RIG TMO CHILE, RIG TMO Colombia where she is an honorary member.

Fernando Jurgensen, MT, has a major in radiology and medical physics and a Bachelor of Radiological Sciences from the University of Tarapaca. He has been a professor of different subjects related to the profession and specialty at Universidad del Desarrollo, Diego Portales University and Andrés Bello University. Currently, he works as a Clinical Applicationist of ultrasound for General Electric. He completed his specialty as a sonographer at the Latin American School of Ultrasound (LASUS) in Miami, USA, obtaining certification in upper and lower limb musculoskeletal ultrasound as well as in musculoskeletal interventions. His research interests are related to the development of the use of diagnostic ultrasound in different areas of clinical imaging, therapeutic support, and intervention.

Juan Guerrero-Henriquez, PT, MSc, currently works as an academic in the Rehabilitation and Human Movement Sciences Department at University of Antofagasta, Chile. He has a master's degree in Orthopedic Manual Therapy from Andrés Bello University, Chile. His research lines are biomechanics, musculoskeletal science, and data science.

ORCID

Juan Guerrero-Henriquez  <http://orcid.org/0000-0002-9071-3586>

References

- [1] Maeda R, Cluff T, Gribble P, et al. Feedforward and feedback control share an internal model of the arm's dynamics. *J Neurosci*. 2018;38(49):10505–10514.
- [2] Myers JB, Lephart SM. The role of the sensorimotor system in the athletic shoulder. *J Athl Train*. 2000;35(3):351–363.
- [3] Riemann B, Lephart S. The sensorimotor system, Part I: The physiologic basis of functional joint stability. *J Athl Train*. 2002;37(1):71–79.
- [4] Winstein C, Wing AM, Whittall J. Chapter 5: Motor control and learning principles for rehabilitation of upper limb movements after brain injury. In: *Handbook of Neuropsychology*. Vol. 9. 2003. p. 77–137. Elsevier Science B.V.
- [5] Fort A, Rodríguez R. Rol del sistema sensoriomotor en la estabilidad articular durante las actividades deportivas. *Apuntes Medicina De l'Esport*. 2013;48(178):69–76.
- [6] Røijezon U, Clark NC, Treleaven J. Proprioception in musculoskeletal rehabilitation: part 1: basic science and principles of assessment and clinical interventions. *Manual Ther*. 2015;20(3):368–377.
- [7] Bateman M, Smith BE, Osborne SE, et al. Physiotherapy treatment for atraumatic recurrent shoulder instability: early results of a specific exercise protocol using pathology-specific outcome measures. *Shoulder and Elbow*. 2015;7(4):282–288.
- [8] Cuéllar R, Ruiz-Ibán MA, Cuéllar A. Anatomy and biomechanics of the unstable shoulder. *Open Orthop J*. 2017;11(1):919–933.
- [9] De la Rosa-Morillo F, Galloza-Otero JC, Micheo W. Rehabilitación del hombro doloroso en el atleta joven. *Rehabilitación*. 2018;1–8. DOI:10.1016/j.rh.2018.10.005
- [10] Di Giacomo G, Piscitelli L, Pugliese M. The role of bone in glenohumeral stability. *Shoulder Elbow*. 2018 December;3(2018):632–640. .
- [11] Desroches G, Desmeules F, Gagnon DH. Characterization of humeral head displacements during dynamic glenohumeral neuromuscular control exercises using quantitative ultrasound imaging: a feasibility study. *Musculoskeletal Sci Pract*. 2016;29:1–5.
- [12] Magarey ME, Jones MA. Dynamic evaluation and early management of altered motor control around the shoulder complex. *Manual Ther*. 2003;8(4):195–206.
- [13] Carreño Mesa F, Osma Rueda J. Diagnóstico de la rotura del manguito de los rotadores (pruebas clínicas e imagenología). Revisión de conceptos actuales. *Revista Colombiana de Ortopedia y Traumatología*. 2016;30(S1):13–25.
- [14] Mackenzie TA, Herrington L, Funk L, et al. Relationship between extrinsic factors and the acromio-humeral distance. *Manual Ther*. 2016;23:1–8.
- [15] Dong W, Goost H, Lin XB, et al. Treatments for shoulder impingement syndrome a prisma systematic review and network meta-analysis. *Medicine (USA)*. 2015;94(10):1–17.
- [16] Whittle S, Buchbinder R. Rotator cuff disease. *Ann Intern Med*. 2015;162(1):ITC1–ITC14.
- [17] Alaiti RK, Harvie DS, Gasparin JT, et al. Distorted distance perception to reachable points in people with chronic shoulder pain. *Musculoskeletal Sci Pract*. 2019;42(April):120–124.
- [18] Lawrence RL, Braman JP, Ludewig PM. Shoulder kinematics impact subacromial proximities: a review of the literature. *Braz J Phys Ther*. 2020;24(3):219–230.
- [19] Warby SA, Watson L, Ford JJ, et al. Multidirectional instability of the glenohumeral joint: etiology, classification, assessment, and management. *J Hand Ther*. 2017;30(2):175–181.
- [20] Beaudreuil J, Lasbleiz S, Aout M, et al. Effect of dynamic humeral centring (DHC) treatment on painful active elevation of the arm in subacromial impingement syndrome. Secondary analysis of data from an RCT. *Br J Sports Med*. 2015;49(5):343–346.
- [21] Benítez-Martínez JC, Casaña-Granell J, Alakhdar-Mohamara Y, et al. Diferencias en el espacio acromio-humeral medido ecográficamente en el dolor de hombro del deportista lanzador. *Fisioterapia*. 2015;37(3):120–127.
- [22] Guerra-Rodríguez D, Rozo L, Basilio D, et al. In vivo measurements of glenohumeral distraction technique performed in three different joint positions. *J Manual Manipulative Ther*. 2021;1–7. DOI:10.1080/10669817.2021.1904350
- [23] Navarro-Ledesma S, Struyf F, Labajos-Manzanares M, Fernandez-Sanchez M, Luque-Suarez A. Is coracohumeral distance associated with pain-function, and shoulder range of movement, in chronic anterior shoulder pain? *BMC Musculoskelet Disord*. 2017;18(136):1–6.
- [24] Savoie A, Mercier C, Desmeules F, et al. Effects of a movement training oriented rehabilitation program on symptoms, functional limitations and acromiohumeral distance in individuals with subacromial pain syndrome. *Manual Ther*. 2015;1–6. DOI:10.1016/j.math.2015.04.004
- [25] Leite MJ, Sá MC, Lopes MJ, et al. Coracohumeral distance and coracoid overlap as predictors of subscapularis and long head of the biceps injuries. *J Shoulder Elbow Surg*. 2019;28(9):1723–1727.
- [26] Kim SY, Weon JH, Jung DY, et al. Effect of the scapula-setting exercise on acromio-humeral distance and scapula muscle activity in patients with subacromial impingement syndrome. *Phys Ther Sport*. 2019;37:99–104.
- [27] Sealey P, Critchley D. Do isometric pull-down exercises increase the acromio-humeral distance? *Physiotherapy (United Kingdom)*. 2017;103(2):208–213.
- [28] Balke M, Banerjee M, Greshake O, et al. The Coracohumeral Distance in Shoulders with Traumatic and Degenerative Subscapularis Tendon Tears. *Am J Sports Med*. 2016;44(1):198–201.
- [29] Zhang H, Zhang Q, Li ZL. Coracohumeral index and coracoglenoid inclination as predictors for different types of degenerative subscapularis tendon tears. *Int Orthop*. 2019;43(8):1909–1916.
- [30] Ager AL, Roy JS, Gamache F, et al. The effectiveness of an upper extremity neuromuscular training program on the shoulder function of military members with a rotator cuff tendinopathy: a pilot randomized controlled trial. *Mil Med*. 2018;1–9. DOI:10.1093/milmed/usy294
- [31] Mottram S, Blandford L. Assessment of movement coordination strategies to inform health of movement and guide retraining interventions. *Musculoskeletal Sci Pract*. 2020 December;45(2019):102100. .
- [32] Spanhove V, De Wandele I, Kjær BH, et al. The effect of five isometric exercises on glenohumeral translations

- in healthy subjects and patients with the hypermobility type of the ehlers-danlos syndrome (heds) or hypermobility spectrum disorder (hsd) with multidirectional shoulder instability: an o. *Physiotherapy*. 2020;107:11–18.
- [33] Holmgren T, Hallgren HB, Öberg B, et al. Effect of specific exercise strategy on need for surgery in patients with subacromial impingement syndrome: randomised controlled study. *BMJ (Online)*. 2012;344(7846):1–9.
- [34] Contemori S, Panichi R, Biscarini A. Effects of scapular retraction/protraction position and scapular elevation on shoulder girdle muscle activity during glenohumeral abduction. *Hum Mov Sci*. 2019 December;64 (2018):55–66. .
- [35] Silva Fernández L, Otón Sánchez T, Fernández Castro M, et al. Maniobras exploratorias del hombro doloroso. *Seminarios de La Fundacion Espanola de Reumatologia*. 2010;11(3):115–121.
- [36] Tennent TD, Beach WR, Meyers JF. A review of the special tests associated with shoulder examination part I: the rotator cuff tests. *Am J Sports Med*. 2003;31(1):154–160.
- [37] Desmeules F, Minville L, Riederer B, et al. Acromiohumeral distance variation measured by ultrasonography and its association with the outcome of rehabilitation for shoulder impingement syndrome. *Clin J Sport Med*. 2004;14(4):197–205.
- [38] Navarro-Ledesma S, Luque-Suarez A. Comparison of acromiohumeral distance in symptomatic and asymptomatic patient shoulders and those of healthy controls. *Clin Biomech*. 2018;53:101–106.
- [39] Faul F, Erdfelder E, Lang AG, et al. G*Power 3: a flexible statistical power analysis program for the social, behavioral, and biomedical sciences. *Behav Res Methods*. 2007;39(2):175–191.
- [40] Seitz AL, Michener LA. Ultrasonographic measures of subacromial space in patients with rotator cuff disease: a systematic review. *Clin Ultrasound*. 2010;39(3):146–154.
- [41] Koo TK, Li MY. A guideline of selecting and reporting intraclass correlation coefficients for reliability research. *J Chiropr Med*. 2016;15(2):155–163.
- [42] Brunkhorst JP, Giphart JE, LaPrade RF, et al. Coracohumeral distances and correlation to arm rotation: an in vivo 3-dimensional biplane fluoroscopy study. *Orthop J Sports Med*. 2013;1(2):1–6.
- [43] Navarro-Ledesma S, Struyf F, Labajos-Manzanares MT, Fernandez-Sanchez M, Morales-Asencio JM, Luque-Suarez A. Does the acromiohumeral distance matter in chronic rotator cuff related shoulder pain? *Musculoskeletal Sci Pract*. 2017;29:38–42.
- [44] Asal N, Şahan MH. Radiological variabilities in subcoracoid impingement: coracoid morphology, coracohumeral distance, coracoglenoid angle, and coracohumeral angle. *Med Sci Monit*. 2018;24:8678–8684.
- [45] Hodges PW. Pain and motor control: from the laboratory to rehabilitation. *J Electromyogr Kinesiol*. 2011;21 (2):220–228.
- [46] Liew BXW, Vecchio AD, Falla D. The influence of musculoskeletal pain disorders on muscle synergies—A systematic review. *PLoS ONE*. 2018;13 (11):1–20.
- [47] Seidi F, Bayattork M, Minoonejad H, et al. Comprehensive corrective exercise program improves alignment, muscle activation and movement pattern of men with upper crossed syndrome: randomized controlled trial. *Sci Rep*. 2020;10(1):1–11.
- [48] Ravichandran H, Janakiraman B, Gelaw AY, et al. Effect of scapular stabilization exercise program in patients with subacromial impingement syndrome: a systematic review. *J Exerc Rehabil*. 2020;16(3):216–226.
- [49] dos Santos C, Jones M, Matias R. Short- and long-term effects of a scapular-focused exercise protocol for patients with shoulder dysfunctions—a prospective cohort. *Sensors*. 2021;21(2888):1–20.
- [50] Salles JI, Velasques B, Cossich V, et al. Strength Training and Shoulder Proprioception. *J Athl Train*. 2015;50 (3):277–280.
- [51] Swanik KA, Lephart SM, Swanik CB, et al. The effects of shoulder plyometric training on proprioception and selected muscle performance characteristics. *J Shoulder Elbow Surg*. 2002;11 (6):579–586.
- [52] de Freitas DG, Marcondes FB, Monteiro RL, et al. Pulsed electromagnetic field and exercises in patients with shoulder impingement syndrome : a randomized, double-blind, placebo-controlled clinical trial. *Arch Phys Med Rehabil*. 2014 Feb;95(2):345–52.DOI:10.1016/j.apmr.2013.09.022. Epub 2013 Oct 15.
- [53] Huang H, Lin J, Leon Y, et al. EMG biofeedback effectiveness to alter muscle activity pattern and scapular kinematics in subjects with and without shoulder impingement. *J Electromyogr Kinesiol*. 2013;23 (1):267–274.
- [54] Larsen CM, Sogaard K, Chreiteh SS, et al. Neuromuscular control of scapula muscles during a voluntary task in subjects with Subacromial Impingement Syndrome . A case-control study. *J Electromyogr Kinesiol*. 2013;1–8. DOI:10.1016/j.jelekin.2013.04.017
- [55] Dingenen B, Blandford L, Comerford M, et al. The assessment of movement health in clinical practice: a multidimensional perspective. *Phys Ther Sport*. 2018;32:282–292.